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1 Title of the Research Project

Machine learning Model to Enhance Business impact of Bangladesh Airlines Services

2 Name of the Project Director (s)

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3 Abstract

The airline industry generates vast volumes of unstructured customer feedback whose systematic analysis can yield a decisive competitive advantage. This study proposes a two-phase text analytics framework that couples sentiment classification with sentiment-aware topic modeling to identify and quantify the business impact of specific airline service dimensions, with the goal of informing service strategy for the rapidly growing Bangladeshi aviation sector. In the first phase, a dedicated data-optimization pipeline combining TF-IDF vectorization, Latent Semantic Analysis, class-wise K-Means clustering with iterative re-clustering, threshold-based sparse-cluster filtering, and SMOTE is introduced to mitigate class imbalance for conventional machine-learning classifiers, while deep-learning and transformers address imbalance natively through inverse-frequency class weighting. A broad range of machine learning, deep learning, and transformer models is benchmarked across three datasets. The optimized Multilayer Perceptron attains the strongest performance (accuracy 0.9004, macro-F1 0.9006, AUROC 0.9683), with tree-based ensembles (Extra Trees, XGBoost, LightGBM) close behind, while substantially outperforming heavier transformer baselines such as RoBERTa at a small fraction of their training cost. In the second phase, the best classifier labels a large multi-carrier review corpus, and a Sentiment-Aware Topic Modeling procedure based on BERTopic extracts coherent themes within each sentiment class. Negative reviews yield the most coherent and best-separated topics (coherence $C_V = 0.4808$; silhouette $= 0.6165$), surfacing recurrent business-risk factors such as flight delays, unprofessional staff, opaque ancillary fees, refund and cancellation handling, and cabin comfort. By converting unstructured reviews into actionable, dimension-level intelligence, the framework offers airline authorities an objective, scalable, and cost-efficient instrument for strategic service improvement.

4 Introduction (including literature survey)

Air travel has become a cornerstone of global mobility and economic integration, with the airline experience a primary component in assessing quality and service. According to the overlaps among fares, route networks, staff support, courtesy, punctuality, cabin comfort, pricing transparency, and the handling of disruptions, these factors are increasingly required to determine customer loyalty, brand reputation, and ultimately revenue of different airline companies. For operating in fast-growing and intense markets, early detection of service weaknesses is essential to enhance service quality. These kinds of analysis are not supplementary; they are a strategic necessity rather than a convenience. As in other regions worldwide, these essential services are especially acute in Bangladesh, where rising disposable incomes, expanding domestic and international demand, and the entry of numerous private operators alongside the national carrier have created a crowded, price-sensitive market. Bangladeshi travelers also regularly fly to a wide range of regional and intercontinental airlines; the competitive benchmark for any local airline is effectively global. In such situations, understanding passengers' satisfaction or dissatisfaction is needed to demonstrate and prioritize operational decisions that directly shape commercial outcomes.

To measure service quality, traditional tools such as periodic satisfaction surveys, interviews, and feedback are used to gauge users' perceptions of airline services. Sezgen et al. (1) identified primary differences in the drivers of passenger satisfaction across cabin class and business model. They assessed staff friendliness and hygiene factors, which are common features used to identify customer satisfaction and dissatisfaction, respectively. Dike et al. (2) systematically assess passengers' expectations and satisfaction in the airline industry using expectation confirmation theory (ECT) and the SERVQUAL framework. It increases understanding of consumer expectations by analyzing passengers. However, these approaches are costly, slow, constrained to predefined attributes, and prone to low response rates and recall bias.

In contrast, feedback on blogs or social media is one way many travelers share their perspectives and experiences with one another, helping others make better-informed decisions about airports and airlines (3). Passengers share a wide range of content on blogs, review platforms, and social media, which is abundant, however, timely, unsolicited, and rich for analysis. These topics are required for mining at scale; however, they pose three challenges. First, sentiment in airline reviews is markedly imbalanced; dissatisfied passengers are far more vocal, so naïve classifiers become biased toward the majority class. Second, the text is short, noisy, and code and punctuation are mixed, which complicates representation learning. Third, raw sentiment labels are insufficient for management, where appropriate decisions require interpretable themes mapped to concrete business dimensions, not merely a positive/negative ratio.

4.1 Related Works

4.1.1 Topic Modeling in Transportation and Service Research

Topic modeling has emerged as a powerful technique for uncovering latent thematic structures within large text corpora. Korfiatis et al. (4) used Structural Topic Models (STM) for airline passengers' online reviews, incorporating document-level covariates via LDA. Results reflect the orthogonality of low-cost as an aspect of airline competition when all other service quality dimensions are considered, thereby explaining the success of low-cost carriers in the airline market. Kherwal and Bhardwaj (5) provided a comprehensive survey and taxonomy of topic modeling methodologies, including an empirical comparison showing that LDA consistently produced more coherent topics than LSA. While the survey is valuable for contextualizing model selection, it acknowledges outstanding challenges

in model stability, visualization, and application across unexplored domains such as text analytics, speech, and image processing. Together, these works establish topic modeling as a mature yet still-evolving instrument and motivate the use of a modern, embedding-based successor (BERTopic) that is better suited to the short, informal text characteristic of online reviews. Sun et al. (6) conducted one of the most comprehensive bibliometric analyses in transportation research by applying LDA to 17,163 articles' abstracts drawn from 22 leading transportation journals spanning 1990 to 2015. The study successfully identified dominant knowledge domains, mapped inter-journal and cross-country similarities, and tracked the temporal evolution of research priorities. The breadth of its coverage and its innovation in quantitative benchmarking make it a foundational reference; however, its reliance on abstracts alone, a fixed number of topics, and a data window that ends in 2015, limits its contemporary applicability.

4.1.2 Sentiment Analysis Frameworks and Methodological Advances

A rich body of literature has examined the comparative performance of sentiment analysis (SA) techniques across three broad paradigms: lexicon-based, traditional machine learning, and deep learning approaches. Liu et al. (7) analyzed airline reviews from TripAdvisor and Skytrax from July 2014 to July 2023 using SERVQUAL theory and employed deep learning models to perform sentiment analysis on airline customer feedback. AlBadani et al. (8) surveyed these paradigms, noting that conventional machine-learning methods such as Support Vector Machines (SVMs) offer high classification accuracy but suffer from slow training times, whereas lexicon-based models are computationally efficient but less accurate. The survey highlights newer transfer-learning architectures, most notably ULMFiT, as high-performance universal solutions that overcome the computational constraints associated with earlier approaches. Samir et al. (9) conducted a systematic review of airline sentiment-analysis studies published between 2018 and 2022, finding that while models achieved varying accuracy levels (68%-98%), most were constrained by small datasets and language-specific applicability. This gap motivated the application of advanced deep-learning architectures, including RNN, LSTM, GRU, CNN, and BERT on larger, more diverse datasets for improved scalability. Similarly, Naresh et al. (10) proposed a hybrid Sequential Minimal Optimization with Decision Tree (SMODT) approach for classifying Twitter data into positive, negative, and neutral categories, achieving 89.47% accuracy. The authors demonstrated that the SMODT framework scales well to large datasets, though they noted that insufficient data led to reduced accuracy in comparable studies. These observations directly inform our emphasis on dataset balancing and on reporting training cost alongside accuracy.

4.1.3 Airline and Aviation Sector Studies

Several studies have applied SA and topic modeling to airline customer reviews to derive actionable insights into service quality. Kwon et al. (11) provided a methodological justification for using topic modeling and sentiment analysis on online airline reviews, arguing that these techniques overcome the limitations of traditional survey instruments by producing objective, scalable, and cost-efficient measurements of customer satisfaction across dimensions, such as in-flight service and staff attitude. During the COVID-19 pandemic, Yu et al. (12) evaluated the passenger satisfaction of AirAsia by combining VADER-based sentiment analysis with LDA topic modeling on 796 online customer reviews. The study yielded a sentiment ratio of approximately 59% positive to 41% negative, with positive feedback concentrated on staff attitude, seat comfort, and negative feedback centered on flight cancellations and protracted refund processes. The authors argue that this combined methodology is a more thorough alternative to conventional surveys, although the relatively small

dataset constrains the generalisability of the findings.

Johari et al. (13) conducted two complementary investigations into Malaysian airline sentiment. First, a lexicon-based VADER labeling approach was combined with supervised classification models, Naïve Bayes (NB), SVM, and a deep learning classifier, applied to reviews of Malaysian Airlines, AirAsia, and Malindo Air. The deep learning approach paired with VADER achieved the highest accuracy of 74.10%, demonstrating the advantage of hybrid lexicon-neural approaches. A subsequent study by the same research group proposed a Long Short-Term Memory (LSTM) ensemble model that analyzes customer reviews with service ratings, where the LSTM-SVM model delivered the strongest predictive performance for customer-recommendation behavior. The limitation of this work is the exclusive focus on English-language content, which may obscure cultural and linguistic nuances. Ahmed et al. (14) investigated airline sentiment from a quantitative ratings perspective, where a multiple-regression model was used to identify labels of review titles that effectively determine the direction and intensity of sentiment. The findings validate the use of simplified title-derived labels as proxies for overall sentiment, though the author recommends future validation across multiple languages and analysis of label co-occurrence patterns. Badanik et al. (15) proposed SASQUE, a sentiment-driven airline service-quality evaluation framework powered by freely available Twitter content and open analytical tools; this framework reduces research costs, provides real-time monitoring, and enables competitive benchmarking, while struggling with double negations, sarcasm, and multi-word phrases. Hasib et al. (3) further reinforced the value of user-generated content (UGC) as a richer and more accessible alternative to formal survey instruments by integrating customer-generated text into a machine-learning pipeline for online airline reviews.

4.1.4 Airport Service Quality Assessment

Research into airport service quality (ASQ) has similarly transitioned from traditional survey instruments to sentiment-driven approaches. Barakat et al. (16) surveyed ASQ evaluation methods and documented the shift away from costly, attribute-constrained surveys toward UGC-based sentiment analysis. It finds that deep learning remains underutilized in this specific aviation sub-domain. Alanazi and Li (17) addressed this gap by applying Aspect-Based Sentiment Analysis (ABSA) alongside a multiclass prediction approach in the airport service reviews, enabling granular evaluation of specific service attributes. Notably, the Random Forest outperformed deep learning models in multiclass prediction, while well-tuned classical models can match or exceed transformers on this classification problem. The authors identify a key limitation of ABSA, such as service aspects must be defined a priori, which favors domain knowledge and may constrain the discovery of emergent themes. Besides, it is precisely the constraint that motivates the unsupervised, sentiment-aware topic modeling.

4.1.5 Broader Transportation, Hospitality, and Travel Contexts

Beyond the airline sector, researchers have applied similar methodologies to adjacent domains. Çaylak (18) analyzed consumer experiences on the Booking.com and Expedia mobile applications using combined sentiment analysis and topic modeling on Google Play reviews; the finding substantially represents higher positive sentiment for Booking.com (81.9%) than Expedia (55.8%), with positive themes centered on ease of booking. Wang et al. (19) examined public sentiment towards the deployment of Baidu's Apollo Go robotaxi service in China, performing sentiment analysis on 36,096 Weibo posts using a fine-tuned hybrid BERT model (accuracy 0.59); enthusiasm was driven by perceptions of technological advancement, while negative sentiment centered on safety concerns and fears of job displacement. Carvache-Franco et al. (20) assessed Twitter as a crisis-communication

channel during COVID-19 for the tourism and hospitality industry, revealing predominantly positive sentiment despite crisis-related terminology, while noting that data collected in early 2020 limits longitudinal transferability.

4.1.6 Summary and Research Gap

The reviewed literature collectively demonstrates three prominent trends. First, there is a clear methodological progression from lexicon-based approaches to hybrid and deep learning models, with ensemble methods consistently achieving superior performance in multiclass and recommendation prediction tasks. Second, UGC from social media and review platforms has largely supplanted or complemented traditional survey instruments, offering greater volume, timeliness, and thematic richness. Third, while studies in related domains (hospitality, autonomous vehicles, tourism) are well established, the specific application of machine learning to assess the business impact of airline service quality spanning operational reliability, comfort, staff conduct, pricing transparency, and customer-loyalty outcomes remains underexplored, and the recurring problem of class imbalance is rarely addressed in a principled way.

The present study addresses this gap by developing a machine-learning framework that integrates sentiment analysis and sentiment-aware topic modeling on airline customer reviews to identify and quantify the business impact of specific service dimensions. By building on the methodological foundations established above, including transformer-based classification, latent-semantic topic extraction, ensemble prediction, and explicitly engineering for class imbalance. This work contributes a domain-specific, decision-oriented analytical framework applicable to strategic airline-service management in Bangladesh and comparable markets.

This work addresses these challenges with a two-phase integrated machine-learning framework. In this model, a data optimization pipeline is proposed in which K-means clustering is employed to balance the input data, and TF-IDF extracts features into an LSA latent space. It performs classwise clustering with iterative re-clustering to prevent any single cluster from dominating, filters sparse clusters, and finally applies SMOTE within the retained clusters. Afterward, various machine learning, deep learning, and transformer architectures are evaluated across three datasets, considering accuracy, precision, recall, macro-F1, AUROC, and training time. This analysis has identified a more accurate computationally economical sentiment classifier. In the second phase, a Sentiment-Aware Topic Modeling (SATM) Framework is proposed, where the best classifier is used to label a large unlabelled corpus. BERTopic is applied separately within each sentiment class, tuned via grid search, and then extracts topics and evaluates their quality using coherence, diversity, and silhouette metrics. In this case, the grid search is used to tune the parameters of different models. The contributions of this work are:

1. **Data Optimization pipeline** This procedure is applied to generate models that manipulate numeric features; sequence and transform models instead receive raw text and a class-weighted loss, avoiding the injection of semantically incoherent synthetic vectors.

2. **SATM Framework.** The best classifier is used to label a large unlabelled corpus, and BERTopic separately applies within each sentiment class and tunes it through grid search, then evaluates topic quality based on coherence, diversity, and silhouette metrics.

3. **Extraction of Business Risk Factors.** The resulting themes are interpreted to isolate the service dimensions most associated with negative sentiment and translate them into strategic recommendations relevant to Bangladeshi carriers.

The remainder of the paper is organized as follows. Section 4.1 reviews related work in topic modeling and sentiment analysis for airline transportation and aviation. Section 5.2 represents a brief description of different procedures that are used in this work. Then, the details of the proposed methodology, evaluation metrics, and datasets are explained. Section 6 reports the experimental results for both sentiment classification and topic modeling. Then, it discusses the findings, their business implications, and limitations, and Section 7 concludes this work and outlines future work about this research.

5 Methodology

5.1 Theoretical Studies of Applied Methods

This section establishes the conceptual foundation underlying the methodologies used in this study. Analyzing complex, unstructured, and imbalanced user-generated text—such as airline service reviews—requires a multi-stage pipeline grounded in established theory. The study draws on Sentiment Analysis and Natural Language Processing (NLP), combining traditional and state-of-the-art architectures. To preserve data integrity, it also incorporates Constrained K-Means Clustering for structured data organization and SMOTE for addressing class imbalance. The classification framework itself is built on a comparative study of Classical Machine Learning, Deep Learning (LSTM, BiLSTM, GRU, CNN), and Transformer-based models (BERT, RoBERTa, DistilBERT) to capture long-range dependencies and nuanced linguistic patterns. The methodology further extends into thematic discovery through BERTopic modeling, grounded in density-based clustering and contextual embeddings, and concludes with Explainable AI (XAI) via SHAP (SHapley Additive exPlanations) to ensure that derived insights remain transparent and actionable for industry decision-making.

The subsections below define the key concepts central to this research, organized from foundational NLP theory through to model architectures and interpretability.

5.1.1 Sentiment Analysis

Sentiment Analysis, often called opinion mining, is a core NLP technique that identifies, extracts, and quantifies subjective opinions expressed in text. While its primary goal is polarity classification (e.g., Positive, Negative, Mixed), this study extends that boundary by addressing the linguistic nuances specific to airline reviews. Theoretically, sentiment analysis has evolved from simple lexicon-based approaches to Machine Learning (ML), Deep Learning (DL), and Transformer-based architectures. This research moves beyond basic TF-IDF vectorization to leverage high-dimensional, contextual word embeddings through models like BERT and RoBERTa, which capture long-range dependencies and bidirectional context essential for interpreting complex passenger feedback. A further theoretical component is the integration of data balancing (SMOTE) and micro-clustering during preprocessing, ensuring a balanced class distribution so that classification models avoid majority-class bias. In the context of service-quality research, sentiment analysis serves as a quantifiable bridge between unstructured human experience and actionable operational intelligence, enabling airlines to pinpoint specific drivers of customer satisfaction.

5.1.2 Feature Extraction

Feature extraction converts raw text into a numerical format that ML and DL models can process. Since airline reviews vary widely in length and complexity, transforming them into weighted feature vectors is essential for capturing sentiment. This research uses two primary representation methods:

- **TF-IDF Vectorization:** Used for classical machine learning models, TF-IDF (Term Frequency–Inverse Document Frequency) assigns weights to words based on their importance within a review relative to the entire dataset, helping models focus on keywords that signal sentiment (e.g., "delay," "comfortable," "average").

- **Contextual Word Embeddings:** Used for Deep Learning and Transformer models (BERT, RoBERTa), these embeddings capture the sequence and relationship between words. For example, in "The flight was not good," the model captures how "not" negates "good," yielding a far more accurate semantic representation than simple word counts.

By combining both techniques, the research captures both the statistical importance of words and their deeper contextual meaning, supporting high-precision sentiment classification. A related count-based approach, GloVe, builds a word co-occurrence matrix and applies dimensionality reduction to produce vectors, offering additional insight into semantic relationships.

5.1.3 Classical Machine Learning for Text Classification

Classical machine learning models provide a strong, interpretable baseline for sentiment classification by establishing decision boundaries based on statistical relationships between features. This research employs a broad range of foundational and ensemble algorithms:

- **Logistic Regression (LR)** (21): Despite its name, a classification method that uses a sigmoid function to map a linear equation's output to a probability, estimating the likelihood that a review belongs to a given sentiment category.

- **Support Vector Machines (SVM) and LinearSVC** (22): SVMs identify the optimal hyperplane that separates sentiment classes with the largest possible margin in high-dimensional feature space. **SVC(RBF)** applies a Radial Basis Function (RBF) kernel to project data into a higher-dimensional space, allowing the model to capture non-linear decision boundaries between sentiment classes. **LinearSVC** is a computationally efficient variant that assumes a linear kernel, making it well suited to the high-dimensional, sparse feature spaces typical of TF-IDF-encoded text.

- **Ridge Classifier** (23): A linear classifier that applies ℓ_2 regularization (a penalty on large coefficients) to the standard least-squares objective, improving generalization and reducing overfitting when many correlated features—such as TF-IDF terms—are present.

- **Stochastic Gradient Descent (SGD)** (24): A linear classifier trained via stochastic gradient descent, updating model weights incrementally on individual samples (or small batches) rather than the full dataset. This makes it highly scalable and efficient for large, high-dimensional text datasets.

- **Perceptron** (25): One of the earliest linear classification algorithms, which iteratively updates a weight vector whenever a training example is misclassified. It forms the conceptual basis for more advanced linear and neural models used elsewhere in this study.

- **Naïve Bayes (NB)** (26): A family of probabilistic classifiers based on Bayes' theorem, assuming conditional independence between features. The Gaussian variant assumes continuous features follow a normal distribution, while the Bernoulli variant models binary/boolean feature representations (e.g., word presence or absence), making it particularly suited to short-text classification tasks.

- **Decision Tree (DT)** (27): A non-parametric model that recursively splits the feature space based on feature-value thresholds that best separate sentiment classes, producing an interpretable, tree-structured set of decision rules.

- **K-Nearest Neighbors (KNN)** (28): A non-parametric, instance-based classifier that assigns a sentiment label based on the majority class among the k closest training samples in feature space, using a distance metric such as Euclidean or cosine similarity.

- **Linear Discriminant Analysis (LDA) and Quadratic Discriminant Analysis (QDA)** (29; 30): Both are generative classifiers that model the feature distribution of each class as Gaussian. LDA assumes all classes share a common covariance structure, yielding linear decision boundaries, whereas **QDA** allows each class its own covariance matrix, producing more flexible quadratic decision boundaries at the cost of requiring more training data.

- **Random Forest (RF)** (31): An ensemble method that constructs multiple decision trees on bootstrapped samples of the data and aggregates their outputs (via majority vote) to improve classification stability and reduce overfitting relative to a single tree.

- **Extremely Randomized Trees (Extra Trees)** (32): Closely related to Random Forest, but introduces additional randomness by selecting split thresholds at random rather than searching for the optimal split at each node. This typically reduces variance further and can improve generalization.

- **Bagging** (33): A general Bootstrap Aggregating that trains multiple instances of a base estimator on different bootstrapped subsets of the training data and combines their predictions, reducing variance and improving robustness to noisy text data.

- **AdaBoost** (34): A boosting ensemble method that sequentially trains weak classifiers, with each successive model focusing more heavily on samples misclassified by its predecessors, ultimately combining them into a single, stronger weighted classifier.

- **Gradient Boosting** (35): An ensemble technique that builds decision trees sequentially, where each new tree is trained to correct the residual errors of the combined ensemble so far, optimizing a differentiable loss function via gradient descent.

- **Extreme Gradient Boosting (XGBoost)** (36): An optimized, regularized implementation of gradient boosting that incorporates second-order gradient information, tree pruning, and parallelized computation, making it highly efficient and widely used for structured and text-based classification tasks.

- **Light Gradient Boosting Machine (LightGBM)** (37): A gradient boosting framework that uses histogram-based algorithms and a leaf-wise (rather than level-wise) tree growth strategy, enabling substantially faster training and lower memory usage than traditional gradient boosting implementations, particularly on large datasets.

- **Multi-Layer Perceptron (MLP)** (38): A feedforward neural network composed of one or more fully connected hidden layers with non-linear activation functions. When applied to TF-IDF or embedding-based feature vectors, the MLP learns non-linear combinations of input features, serving as a bridge between classical feature-based models and more complex sequence-aware deep learning architectures.

These models are particularly effective in this study because they leverage well-engineered TF-IDF features (39) to learn patterns that distinguish positive, negative, and mixed feedback in aviation reviews, while the ensemble and boosting variants further improve robustness to noisy or imbalanced review text.

5.1.4 Deep Learning Architectures

Deep Learning is a subset of machine learning that uses hierarchical neural networks to process data in a manner loosely inspired by the human brain (38). Unlike classical algorithms, it excels at handling large volumes of unstructured text and extracting complex patterns—critical for analyzing thousands of airline reviews. Deep learning also demands substantial computational power, with the parallel architecture of modern GPUs well suited to this workload (9).

This study uses several deep learning architectures to capture the sequential and structural nature of language, where classical models often treat words as independent units:

- **Long Short-Term Memory (LSTM)** (40): Captures word-level context across long sequences, addressing the vanishing gradient problem so that early words (e.g., a "not") can still influence sentiment interpreted later in a review.
- **Bidirectional LSTM (BiLSTM)** (41): Extends LSTM by processing sequences in both forward and backward directions, combining both representations to capture context from past *and* future words in a sentence.
- **Gated Recurrent Unit (GRU)** (42): A simpler, often faster alternative to LSTM that uses reset and update gates to regulate information flow while still mitigating the vanishing gradient problem.
- **Convolutional Neural Network (CNN)** (43; 44): Identifies local patterns and key phrases within text through layers of interconnected nodes that automatically learn feature representations without manual intervention.

Together, these architectures—combined with large labeled datasets and sufficient computational power—enable high-precision classification of passenger sentiment into Positive, Negative, and Mixed categories.

5.1.5 Transformer Architecture

The Transformer architecture (45) represents a major shift in NLP, moving away from sequential word-by-word processing toward a holistic, parallel approach. At its core is the self-attention mechanism, which allows the model to weigh every word in a sentence simultaneously. For example, in "The flight was long, but the service was great," self-attention connects "great" to "service" and "long" to "flight," regardless of their distance in the sentence. Unlike LSTM or GRU's sequential processing, Transformers process an entire sequence at once, making training faster and preventing the loss of information common in long sentences. To preserve word order despite this parallelism, the architecture uses positional encoding, which tags each word with information about its position in the sequence. This research uses the following Transformer-based models:

- **BERT / RoBERTa** (46; 47): Bidirectional models that read text left-to-right and right-to-left simultaneously, capturing deep contextual understanding. RoBERTa is a robustly optimized variant of BERT with an improved training procedure, achieving stronger performance on downstream NLP tasks.

- **DistilBERT** (48): A smaller, faster, and lighter version of BERT, produced via knowledge distillation, that retains most of BERT's performance while reducing size and computational cost.
- **ALBERT (A Lite BERT)** (49): A parameter-efficient variant of BERT that uses factorized embedding parameterization and cross-layer parameter sharing to substantially reduce model size and memory footprint, while maintaining competitive performance on language understanding tasks.

This bidirectional, parallel design makes Transformers significantly more capable than classical ML or standard DL models at identifying the subtle, fine-grained sentiments found in complex passenger feedback.

5.1.6 Topic Modeling

Topic Modeling is an unsupervised technique for uncovering hidden semantic structures, or "topics," within large collections of unstructured text such as airline reviews. While traditional methods like Latent Dirichlet Allocation (LDA) (50) rely on probabilistic mixtures of words and documents, this research uses BERTopic (51), a more modern framework that produces semantically coherent results without requiring pre-labeled training data.

BERTopic identifies underlying themes—such as "Operational & Airport Issues" or "General Positive Experience"—through the following process: it first creates dense document representations using contextual embeddings (e.g., Sentence-BERT) (52), then reduces their dimensionality using Uniform Manifold Approximation and Projection (UMAP) (53) before grouping these representations using the density-based clustering algorithm HDBSCAN (54) to find related reviews. Finally, a class-based TF-IDF (c-TF-IDF) technique (51) extracts the most representative words for each cluster, ensuring the resulting topics are distinct and interpretable. This methodology bridges broad sentiment scores and specific service concerns, enabling a granular understanding of which operational areas drive customer satisfaction or dissatisfaction.

5.2 Proposed Methodology

This study proposes a comprehensive text analytics framework that uses an effective machine-learning model for predicting passenger sentiment toward airline services, while uncovering the key factors needed to inform business strategy for the airline industry in Bangladesh. The framework comprises two main phases, illustrated in Figure 1. At first, a sentiment-classification module employs a range of Natural Language Processing (NLP) and state-of-the-art classification techniques to analyze and evaluate high-performing sentiment-prediction models. In the second phase, topic-modeling techniques uncover the underlying thematic structure of airline-service reviews, identify potential risk factors, and generate actionable business insights through comprehensive visualizations.

5.2.1 Data Preparation

The study first collects relevant airline customer-review datasets from public repositories and prior literature. Multiple datasets are integrated to provide a more comprehensive representation of airline-service experiences. Then, a series of NLP operations is applied to the review texts, including text normalization, punctuation removal, stop-word removal, and lemmatization. These steps

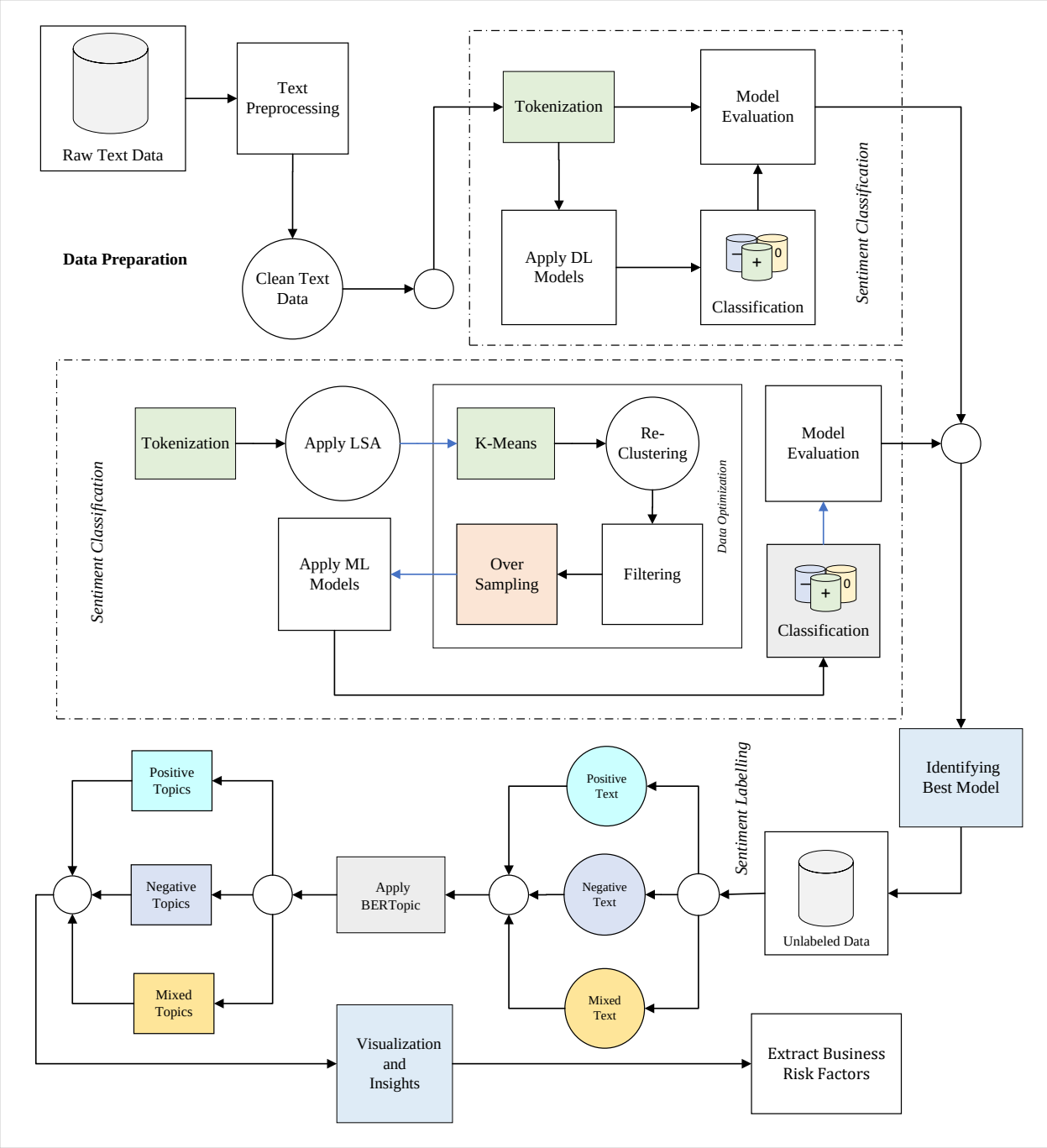


Figure 1: Proposed two-phase framework: sentiment classification followed by sentiment-aware topic modeling and business-risk extraction.

eliminate noise, improve text quality, and produce a clean, standardized corpus suitable for the subsequent analyses.

5.2.2 Data Optimization

A series of techniques is employed to optimize the input data when the dataset exhibits class imbalance and related complexity. The pipeline consists of dimensionality reduction, class-wise K-Means clustering, iterative re-clustering, threshold-based filtering, SMOTE-based oversampling, and cluster-to-label remapping (see Algorithm 1). Due to the raw text cannot be processed directly by distance-based clustering such as K-Means, each preprocessed review is first transformed into a high-dimensional sparse vector using TF-IDF weighting over a vocabulary of $V = 1{,}000$ terms:

$$\mathrm{TF-IDF}(t, d) = \mathrm{tf}(t, d) \times \log \frac{N}{1 + \mathrm{df}(t)} \tag{1}$$

where $\mathrm{tf}(t, d)$ is the frequency of term t in document d, N is the total number of documents, and $\mathrm{df}(t)$ is the number of documents containing t. The resulting sparse matrix $X \in \mathbb{R}^{n \times V}$ is projected into a dense, low-dimensional latent semantic space via Truncated Singular Value Decomposition (SVD), LSA:

$$X \approx U_r \Sigma_r V_r^\top, \qquad X_{\mathrm{red}} = U_r \Sigma_r \in \mathbb{R}^{n \times r} \tag{2}$$

Afterward, K-Means is applied separately within each sentiment class rather than over the entire corpus. For each sentiment subset X_s, K-Means partitions the data into k_s clusters by minimizing the within-cluster sum of squares (WCSS):

$$\arg \min_{\{C_1, \ldots, C_{k_s}\}} \sum_{j=1}^{k_s} \sum_{x_i \in C_j} \|x_i - \mu_j\|^2 \tag{3}$$

where $\mu_j = \frac{1}{|C_j|} \sum_{x_i \in C_j} x_i$ is the centroid of cluster j. Each sample is assigned to its nearest centroid:

$$z_i = \arg \min_{j \in \{1, \ldots, k_s\}} \|x_i - \mu_j\|^2 \tag{4}$$

The optimal k_s is selected per class by maximizing the Silhouette Score, which measures how similar a sample is to its own cluster relative to the nearest neighboring cluster. Following the initial clustering, inter-cluster disparities are substantial. If one cluster $C_{\max}$ greatly exceeds the others, SMOTE is used to generate an excessive number of synthetic samples to equalize it. The expected number of synthetic samples SMOTE is generated by:

$$S_{\mathrm{synthetic}} = \sum_{j=1}^{K} \left(\max_l |C_l| - |C_j| \right) \tag{5}$$

where K is the total number of clusters across all sentiment classes. To minimize $S_{\mathrm{synthetic}}$, this algorithm iteratively splits the largest cluster. At each iteration t, the largest cluster $C_{\max}^{(t)}$ is identified, and the number of sub-clusters is computed as:

$$k^* = \left\lceil \frac{|C_{\max}^{(t)}|}{|C_{\mathrm{second}}^{(t)}|} \right\rceil \tag{6}$$

where $|C_{\mathrm{second}}^{(t)}|$ is the size of the second-largest cluster. K-Means is applied to $C_{\max}^{(t)}$ for $k \in \{k^*, k^*+1, k^*+2\}$, where the split with the highest Silhouette Score is accepted. Finally, the iteration terminates when $|C_{\max}^{(t)}| \leq |C_{\mathrm{second}}^{(t)}|$ or no valid split is found. Very small clusters are then removed,

as they contain insufficient intra-class diversity to support meaningful SMOTE interpolation; the filtering threshold is:

$$\tau = \max(\delta_{\min},\ \lfloor \alpha \cdot \tilde{n}_C \rfloor) \tag{7}$$

After filtering, SMOTE equalizes the retained clusters. For each minority-cluster sample x_i, a synthetic sample is generated by linear interpolation toward one of its $k_{nn} = 3$ nearest neighbors $x_{i(j)}$:

$$\tilde{x} = x_i + \lambda \cdot \left(x_{i(j)} - x_i\right), \qquad \lambda \sim U(0,1) \tag{8}$$

After balancing, each cluster label is remapped to its original sentiment label based on the sentiment prefix of the cluster name, yielding the final SMOTE-balanced dataset $(X_{\text{bal}}, y_{\text{bal}})$ used to train conventional ML classifiers.

This optimization is applied exclusively to models that consume numeric feature representations. Deep learning and transformers are excluded because synthetic SMOTE vectors exist only in the low-dimensional LSA embedding space, which contains no recoverable linguistic content; feeding them into sequence models or subword tokenizers would yield semantically incoherent inputs. Instead, these models receive the original, unmodified text and address class imbalance natively through inverse-frequency class weights applied to the cross-entropy loss:

$$\mathcal{L} = -\sum_{i=1}^{n} w_{y_i} \log p(y_i \mid x_i), \qquad w_c = \frac{n}{C \cdot n_c} \tag{9}$$

where n is the total number of training samples, C is the number of classes, n_c is the count of class c, and $p(y_i \mid x_i)$ is the predicted probability of the true class.

5.2.3 Sentiment Classification

To enable a robust comparative analysis and to identify the most effective model, the study evaluates a broad set of classifiers spanning three paradigms:

- **Classical machine learning:** LR, Ridge Classifier, SGD, Perceptron, LDA, QDA, GNB, BNB, KNN, DT, RF, ET, Bagging, AdaBoost, XGBoost, LGBM, SVM, Linear SVC, and MLP.
- **Deep learning:** LSTM, BiLSTM, GRU, and CNN.
- **Transformers:** BERT, DistilBERT, RoBERTa, and ALBERT.

All classical models consume the SMOTE-balanced LSA features produced by Algorithm 1, whereas deep learning and transformer models consume raw text with class-weighted loss (Eq. 9). Model performance is evaluated using established classification metrics (Section 5.3.1), and the best-performing model is retained with its fitted preprocessing artifacts for the topic-modeling stage. The ten most accurate and stable models are reported in Section 6.

5.3 Topic Modeling

In this phase, the best classifier assigns sentiment labels to a large unlabelled corpus of airline reviews from a publicly accessible repository and predicts confidence scores, which are recorded to assess classification reliability. Granular Topic Modeling (BERTopic) is then applied to discover latent themes within the reviews, with each sentiment category (i.e., positive, negative, and neutral/mixed)

Algorithm 1 Data Optimization Framework

Require: Raw dataset D; min. cluster-size floor $\delta_{\min}$; cluster-size ratio α; neighbour count k_{nn}; LSA components r; vocabulary size V
Ensure: Noise-free, well-distributed dataset D_{balanced}

procedure DATAOPT($D, \delta_{\min}, \alpha, k_{nn}, r, V$)
  Compute TF–IDF matrix $X \in \mathbb{R}^{n \times V}$ from D
  Apply LSA: $X_{\text{red}} \leftarrow U_r \Sigma_r \in \mathbb{R}^{n \times r}$
  Partition D into class subsets $\{C_1, \ldots, C_K\}$ by sentiment label
  Initialize $D_{\text{balanced}} \leftarrow \emptyset$
  for each class subset C_k **do**
    Select optimal k_s by maximising $s(i)$ over $k \in \{2, \ldots, 10\}$ ▷ silhouette-based k
    $\Psi_k \leftarrow \text{KMEANS}(X_{\text{red}}[C_k], k_s)$ ▷ initial clustering
    while $|C^{(t)}_{\max}| > |C^{(t)}_{\text{second}}|$ **do**
      $k^* \leftarrow \lceil |C^{(t)}_{\max}| / |C^{(t)}_{\text{second}}| \rceil$ ▷ recursive re-clustering
      for each $\psi_j \in \Psi_k$ with $|\psi_j| = |C^{(t)}_{\max}|$ **do**
        $\Psi_k \leftarrow (\Psi_k \setminus \{\psi_j\}) \cup \text{KMEANS}(\psi_j, k^*)$
      end for
    end while
    $\Psi_{\text{valid}} \leftarrow \{\psi_j \in \Psi_k : |\psi_j| \geq \max(\delta_{\min}, \lfloor \alpha \, \tilde{n}_C \rfloor)\}$ ▷ sparse-cluster elimination
    for each $\psi_j \in \Psi_{\text{valid}}$ **do**
      $\tilde{\psi}_j \leftarrow \text{SMOTE}(\psi_j, k_{nn})$ ▷ oversample minority clusters
      $\psi_j \leftarrow \psi_j \cup \tilde{\psi}_j$
    end for
    $D_k \leftarrow \bigcup_{\psi_j \in \Psi_{\text{valid}}} \psi_j$ ▷ merge within class
    $D_{\text{balanced}} \leftarrow D_{\text{balanced}} \cup D_k$ ▷ merge across classes
  end for
  return D_{balanced}
end procedure

treated separately to surface the themes specific to each, following the Sentiment-Aware Topic Modeling (SATM) procedure (Algorithm 2).

For each sentiment subset D_k, documents are embedded with a SentenceTransformer (`all-MiniLM-L6-v2`); dimensionality is reduced with UMAP (with neighbor count $n_{nb} = \max(10, \lfloor |D_k|^{0.5} \rfloor)$); clusters are formed with HDBSCAN under a minimum cluster size δ; and cTF-IDF yields the top-η representative terms per topic. BERTopic hyperparameters θ^*_k are selected per class by grid search to maximize topic coherence. Topic quality is then quantified with coherence C_V, topic diversity, and silhouette (Section 5.3.1), and a suite of visualizations, including word clouds, cTF-IDF bar charts, and inter-topic distance maps, and topic hierarchies, which support interpretation. The resulting per-class topics, weighted by document share, are compiled into an insight table from which business-risk factors are extracted.

5.3.1 Evaluation Metrics

The effectiveness of the trained models is assessed using several evaluation metrics that quantify their ability to detect sentiments of airline services. A confusion matrix is used to extract TP, TN, FP, and FN, which denote true positives, true negatives, false positives, and false negatives,

Algorithm 2 Sentiment-Aware Topic Modeling (SATM) Framework

Require: Unlabelled corpus D; frozen artifacts $\{T, L, S, f^*\}$ of best classifier; grid space G; min. cluster size δ; top-η words

Ensure: Topic sets $\{\Phi_k\}$; metrics $\{C_V^{(k)}, \mathrm{Div}^{(k)}, \mathrm{Sil}^{(k)}\}$; insight table I; report R

procedure SATM($D, T, L, S, f^*, G, \delta, \eta$)
  $D_{\text{clean}} \leftarrow \textsc{Preprocess}(D)$
  $X \leftarrow S(L(T(D_{\text{clean}})))$ ▷ TF–IDF → LSA → Scaler
  $\hat{y}_i \leftarrow \arg\max_c f^*.\textsc{PredictProba}(X)_{ic}$, $\hat{s}_i \leftarrow P_{i,\hat{y}_i}$
  $D_k \leftarrow \{d \in D_{\text{clean}} : \hat{y}_d = k\}$, $k \in \{\text{Pos}, \text{Neg}, \text{Mix}\}$
  $E \leftarrow \textsc{SentenceTransformer}(\texttt{all-MiniLM-L6-v2})$
  for each sentiment class k **do**
    $\theta_k^* \leftarrow \arg\max_{\theta \in G} C_V(\text{BERTopic}_\theta, D_k)$ ▷ grid search
    $n_{nb} \leftarrow \max(10, \lfloor |D_k|^{0.5} \rfloor)$
    $M_k \leftarrow \text{BERTopic}(E, \text{UMAP}(\theta_k^*, n_{nb}), \text{HDBSCAN}(\delta), \eta)$
    $\Phi_k \leftarrow M_k.\textsc{FitTransform}(D_k)$
  end for
  for each sentiment class k **do**
    $C_V^{(k)} \leftarrow \textsc{CoherenceModel}(\Phi_k, D_k)$
    $\mathrm{Div}^{(k)} \leftarrow |\bigcup_j W_j^{(k)}| / \sum_j |W_j^{(k)}|$
    $\mathrm{Sil}^{(k)} \leftarrow \textsc{SilhouetteScore}(E(D_k), \Phi_k)$
  end for
  for each sentiment class k **do**
    $\textsc{PlotWordCloud}(M_k, \eta)$; $\textsc{PlotTopWords}(M_k, \eta)$
    $\textsc{PlotMDS}(M_k, E)$; $\textsc{PlotHierarchy}(M_k, D_k)$
    $I_k \leftarrow \{(\phi_j, W_j, |\phi_j| / \sum_{j'} |\phi_{j'}|)\}$ ▷ top-N topics
  end for
  $I \leftarrow \bigcup_k I_k$; $R \leftarrow \textsc{GenerateReport}(I, \{C_V^{(k)}, \mathrm{Div}^{(k)}, \mathrm{Sil}^{(k)}\})$
  return $\{\Phi_k\}$, $\{C_V^{(k)}, \mathrm{Div}^{(k)}, \mathrm{Sil}^{(k)}\}$, I, R
end procedure

respectively. This task is multiclass and imbalanced, and the overall accuracy is reported alongside precision, recall, F1-Score, AUROC, and other measures to provide a comprehensive understanding of the model's strengths and areas for improvement. In this case, the weights for every class are equal, thereby penalizing majority-class bias. The following metrics are described below:

- **Accuracy** measures the overall proportion of correct predictions made by the model. It is calculated as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{10}$$

- **Precision** is a key metric that measures the accuracy of the model when predicting positive instances, which is calculated as:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{11}$$

- **Recall**, also known as Sensitivity or True Positive Rate, measures the model's ability to correctly identify all actual positive instances.

$$\text{Recall} = \frac{TP}{TP + FN} \tag{12}$$

- **F1-Score** is the harmonic mean of precision and recall. It provides a balanced measure of a model's performance, especially useful when there is a need to account for both false positives and false negatives. The formula is:

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{13}$$

$$\text{Macro-F1} = \frac{1}{C} \sum_{c=1}^{C} \text{F1}_c \tag{14}$$

We additionally represent the macro one-vs-rest Area Under the ROC curve (AUROC), which summarizes ranking quality across decision thresholds and is robust to class imbalance, and **training time (s)** as a measure of computational cost.

5.4 Topic modeling

Topic quality is assessed with three complementary measures:

- **Coherence (C_V).** combines normalized pointwise mutual information with cosine similarity over a sliding window to quantify the semantic relatedness of a topic's top terms; higher values indicate more interpretable topics.
- **Topic diversity** is the proportion of unique words among the aggregated top-η words across all topics, $\text{Div} = \left|\bigcup_j W_j\right| / \sum_j |W_j|$; values near 1 indicate minimal redundancy between topics.
- **Silhouette score** measures how well each document embedding fits its assigned topic cluster relative to the nearest competing cluster, indicating cluster separation and compactness.

5.5 Dataset Descriptions

Three datasets are used for sentiment classification, and one large corpus is used for topic modeling. Each review is labeled into one of three sentiment classes: **positive**, **negative**, and **mixed/neutral**.

- **D1 balanced (standard pipeline).** This dataset contains structured customer feedback and sentiment data evaluating multiple airlines operating in Bangladesh. The data is primarily derived from online travel reviews and is designed for researchers investigating customer satisfaction, sentiment analysis, hospitality management, and service quality in regional aviation sectors. To evaluate customer satisfaction and service quality within the Bangladeshi aviation industry, an initial core dataset comprising 1,047 unique passenger reviews was compiled and analyzed. There are 4 columns: reviews, target, title, and airlines. The dataset was streamlined to include only two primary columns: the raw review narrative and the target sentiment label. The auxiliary columns for the specific 'airline' and the 'review' title were excluded from the analysis as they were deemed irrelevant to the current research objectives. The target variable uses a multi-class

sentiment schema, evenly split into Positive, Negative, and Mixed, with exactly 349 instances per class. The explicit quantitative breakdown of the corpus is structured in Table 1.

A class-balanced dataset processed through a standard classification pipeline (TF-IDF → LSA, without the proposed re-clustering and SMOTE optimization). D1 serves as a baseline, isolating model behavior in the absence of class imbalance.

- **D2 imbalanced (proposed pipeline).** Dataset-2 comprises 7,078 total reviews distributed across three classes: Negative (49.36%), Positive (28.78%), and Mixed (21.86%). The majority-to-minority class imbalance ratio stands at 2.26. According to the standard evaluation thresholds, this indicates a mild imbalance (within the 1–3x range), confirming that the dataset maintains a relatively healthy distribution across categories for robust model training.

- **D3 imbalanced (proposed pipeline).** The final target dataset (Dataset-3) is systematically constructed by integrating two distinct source datasets: Dataset-1 and Dataset-2. The first repository, Dataset-1, contains 1,047 unique reviews. The second repository, Dataset-2, encompasses 7,078 reviews. By concatenating these two data structures, a comprehensive consolidated dataset consisting of exactly 8,125 reviews was generated for the multi-class sentiment analysis framework.

  Two larger, naturally imbalanced corpora that reflect the real-world tendency for dissatisfied passengers to review more frequently are processed by the proposed data-optimization pipeline in Algorithm 1. D2 and D3 differ in source composition and scale, allowing the robustness of the optimization to be assessed across distinct distributions.

Table 1: Sentiment Class Distribution Across Datasets

Dataset	Sentiment Class	Review Count	Percentage (%)
D1	Positive	349	33.33
	Negative	349	33.33
	Mixed	349	33.33
D2	Positive	3,494	49.36
	Negative	2,037	28.78
	Mixed	1,547	21.86
D3	Positive	3,843	47.30
	Negative	2,386	29.37
	Mixed	1,896	23.34

For the topic-modeling phase, the best classifier was applied to a large multi-carrier corpus of online airline reviews and partitioned by predicted sentiment. The resulting subsets yielded 110 positive, 127 negative, and 215 mixed topics (Table 3); the ten largest topics within each sentiment class are summarized in Tables 4 and 6 as well as visualized in Figures 2, 3, 6, and 7, respectively.

6 Results and discussion

This section presents the empirical findings of the two-phase framework, in which different datasets (i.e., D1, D2, and D3) were analyzed employing various machine learning models. Section 6.1

outlines the experimental protocol; Section 6.2 presents the sentiment-classification benchmark and Section 6.3 isolates the effect of the data-optimization pipeline; Section 6.4 evaluates topic-model quality and Section 6.5 interprets the thematic structure of each sentiment class; and Section 6.6 consolidates the extracted business-risk factors.

6.1 Experimental Setup

All models were trained and evaluated under a consistent protocol. Each labeled dataset was split into stratified training and test partitions to preserve the original class proportions in the held-out set so that evaluation reflects realistic deployment conditions. Different classifiers were trained on the SMOTE-balanced LSA features (Algorithm 1), with the balancing applied to the training partition to prevent information leakage into the test set; deep-learning and transformer models received raw text with the inverse-frequency class-weighted loss of Eq. (9). For topic modeling, BERTopic hyperparameters were tuned per sentiment class by coherence-maximizing grid search (Algorithm 2). Classification quality is reported using accuracy, macro-F1, and macro AUC, and computational cost using wall-clock training time; topic quality is reported using coherence (C_V), diversity, and silhouette.

6.2 Sentiment Classification Performance

Table 2 reports the ten most accurate and stable classifiers on each dataset. On the balanced baseline **D1**, the **Extra Trees (ET)** ensemble is clearly the strongest model, attaining an accuracy of 0.8524, macro-F1 of 0.8520, and AUROC of 0.9455, ahead of Random Forest (0.8095) and the remaining models, which cluster in the 0.71–0.76 accuracy range. The Support Vector Classifier (SVC) performs poorly on D1 (0.5762 accuracy), and notably, the CNN (0.7571) and RoBERTa (0.7476) do not surpass the tree-based ensemble despite their greater representational capacity.

On the two imbalanced datasets in the proposed pipeline, classifier performance improves substantially, with the Multilayer Perceptron (MLP) emerging as the best overall model. On **D2**, the MLP achieves an accuracy of 0.8923 and macro-F1 of 0.8917, with XGBoost (0.8839), Extra Trees (0.8815), and LightGBM (0.8787) close behind; Extra Trees (ET) attains the highest AUC (0.9741). On **D3**, the MLP again leads with an accuracy of 0.9004, macro-F1 of 0.9006, and AUC of 0.9683, followed by SVC (0.8899), XGBoost (0.8867), and LightGBM (0.8853). Across both imbalanced datasets, the optimized classical and shallow neural models occupy the entire top tier, while the transformer (RoBERTa, ≈ 0.85) and CNN (≈ 0.76) baselines trail by four to fourteen percentage points.

A second pattern concerns computational economy. The best models are also among the cheapest to train: the winning MLP requires roughly 9-10 s on D2/D3, and ET about 6 s, whereas RoBERTa consumes 375.6 s on D2 and 427.8 s on D3, an order-of-magnitude more compute for materially lower accuracy. Bagging is similarly expensive (84-151 s) but yields no corresponding increase in accuracy. The optimized MLP, therefore, offers the most favorable accuracy-to-cost ratio and is selected as the best model and carried forward to the topic-modeling phase.

6.3 Effect of the Data-Optimization Pipeline

The benchmark provides consistent evidence that the proposed optimization pipeline is effective on imbalanced data. Although D1 (balanced, standard pipeline) is a distinct corpus from D2 and D3, the imbalanced datasets processed through Algorithm 1 nonetheless yield higher best-model accuracy (0.8923 and 0.9004) than the balanced baseline (0.8524), and more importantly they

Table 2: Performance comparison of the top-10 stable classification models across datasets. The best value per metric within each dataset is **bold**.

Dataset	Model	Accuracy	F1	AUROC	Train (s)
D1 Balanced (standard pipeline)	ET	**0.8524**	**0.8520**	**0.9455**	5.60
	RF	0.8095	0.8085	0.9281	2.62
	LGBM	0.7571	0.7573	0.8773	0.87
	MLP	0.7286	0.7253	0.8977	4.71
	XGB	0.7476	0.7483	0.8768	8.06
	RoBERTa	0.7476	0.7421	0.8962	0.56
	Bagging	0.7095	0.7108	0.8857	11.74
	SVC	0.5762	0.5522	0.8059	9.50
	LR	0.7143	0.7133	0.9176	0.56
	CNN	0.7571	0.7570	0.9027	6.00
D2 Imbalanced (proposed pipeline)	MLP	**0.8923**	**0.8917**	0.9612	9.02
	ET	0.8815	0.8776	**0.9741**	6.02
	XGB	0.8839	0.8820	0.9665	13.97
	LGBM	0.8787	0.8768	0.9641	9.79
	SVC	0.8750	0.8721	0.9657	22.15
	RF	0.8619	0.8575	0.9651	31.97
	Bagging	0.8470	0.8433	0.9562	83.75
	RoBERTa	0.8453	0.8414	0.9501	375.60
	LR	0.7901	0.7858	0.9188	0.39
	CNN	0.7599	0.7665	0.8984	9.00
D3 Imbalanced (proposed pipeline)	MLP	**0.9004**	**0.9006**	0.9683	10.41
	SVC	0.8899	0.8881	**0.9720**	30.88
	XGB	0.8867	0.8853	0.9701	16.92
	LGBM	0.8853	0.8837	0.9676	9.21
	ET	0.8791	0.8748	0.9705	5.79
	RF	0.8683	0.8644	0.9605	44.65
	Bagging	0.8511	0.8479	0.9503	151.30
	RoBERTa	0.8498	0.8491	0.9498	427.80
	LR	0.8122	0.8080	0.9300	2.02
	CNN	0.7600	0.7638	0.9036	9.00

raise the entire upper tier: on D2 and D3 the top six to seven models all exceed 0.86 accuracy and 0.95 AUROC, whereas on D1 only the leading ensemble clears 0.85. This compression of the leaderboard toward high accuracy indicates that the class-wise clustering, recursive splitting of dominant clusters, sparse-cluster filtering, and within-cluster SMOTE jointly produce a cleaner, better-conditioned feature space in which most numeric-feature models perform strongly. The high macro-F1 values (which track accuracy closely throughout Table 2) confirm that these gains are not an artifact of majority-class bias, however, reflect balanced per-class performance.

The pipeline also clarifies where model capacity is most helpful. Because synthetic SMOTE vectors are confined to the LSA space and are withheld from sequence and transformer models, the latter rely solely on class-weighted loss to cope with imbalance, an approach that, on the present

corpora, leaves them below the optimized classical models. This supports the design decision to balance numeric-feature models with SMOTE and text models with loss weighting, rather than forcing a single strategy across heterogeneous architectures.

6.4 Topic Modeling Quality

Using the best classifier (the optimized MLP) to label the unlabeled corpus, the SATM procedure produced 110, 127, and 215 topics for the positive, negative, and mixed classes, respectively.

Table 3 reports the quality metrics. Negative reviews yield the most coherent and best-separated topics (coherence $C_V = 0.4808$; silhouette = 0.6165), indicating that complaints cluster tightly around a small number of well-defined issues. Mixed reviews are the most diverse (Div = 0.6112) and the most numerous (215 topics), consistent with their semantically heterogeneous, ambivalent content. Positive reviews show moderate coherence (0.3784) and the highest internal overlap (lowest diversity, 0.5964), reflecting the tendency of satisfied passengers to use a common, repetitive vocabulary of praise (e.g. *good*, *service*, *seat*, *food*) across many carriers. The absolute coherence values (0.37–0.48) are typical for short-text, embedding-based topic modeling on social and review data and are adequate for the interpretive purpose of this study.

Table 3: Topic-model quality metrics by sentiment category.

Sentiment	**Coherence (C_V)**	**Diversity**	**Silhouette**	**# Topics**
Positive	0.3784	0.5964	0.5736	110
Negative	**0.4808**	0.5362	**0.6165**	127
Mixed	0.3664	**0.6112**	0.6125	215

6.5 Thematic Findings per Sentiment Class

Positive topics (Table 4; Figures 2 and 3). The ten largest positive topics are organized primarily around specific carriers and routes, accompanied by recurring praise terms. They include Air Canada Rouge / Toronto (T-P1, 847 docs), Thai Airways / Bangkok (T-P2, 836), Qantas / Sydney-Brisbane (T-P3, 661), Qatar Airways / Doha (T-P4, 509), Lufthansa / Frankfurt-Munich (T-P5, 503), China Southern / Guangzhou (T-P6, 458), Turkish Airlines / Istanbul (T-P7, 407), British Airways / Heathrow (T-P8, 378), Air France / Paris-CDG (T-P9, 359), and JFK / New York routings (T-P10, 353). The cTF-IDF profiles show that positive sentiment attaches to premium full-service carriers and is expressed through cabin and catering vocabulary (*seat*, *food*, *business*, *lounge*, *service*), identifying comfort and on-board service as the principal drivers of satisfaction.

Negative topics (Table 5; Figures 4 and 5). The negative themes are the most decision-relevant. They are dominated by United Airlines / Houston-Newark (T-N1, 1,014 docs) and a carrier-agnostic delay cluster (T-N2, 817: *delayed*, *delay*, *hour*, *minute*, *gate*), followed by Mexican carriers (T-N3, 683), Ryanair / Dublin (T-N4, 664: *print*, *boarding*, *pay*, *euro*, *priority* i.e. ancillary-fee friction), Air Canada Rouge (T-N5,633: *entertainment*, *seat*, *ipad*, *cramped* comfort and IFE complaints), Etihad / Abu Dhabi (T-N6, 633), Delta / Atlanta (T-N7, 515: again *delayed*, *hour*), Southwest (T-N8, 488: *customer*, *last*, *get*—service-recovery friction), a carrier-agnostic **staff-conduct** cluster (T-N9, 482: *rude*, *attendant*, *staff*, *attitude*, *unprofessional*), and Tigerair / Virgin Australia (T-N10, 482). Two of the largest negative clusters (delays and rude staff) are **cross-carrier**, signaling systemic rather than carrier-specific risks.

Mixed topics (Table 6; Figure 6 and 7). The mixed class captures ambivalent experiences, concentrated on premium Asian and European carriers and hubs: Qatar / Doha (T-M1, 459: *amazing*, *best*, *service*), Air France / Paris (T-M2, 432), China Southern / Guangzhou (T-M3, 429), Pacific hubs LAX-SFO-ICN (T-M4, 348), Aegean / Athens (T-M5, 330), KLM / Amsterdam (T-M6, 302), Vietnam carriers / Hanoi (T-M7, 293), ANA-JAL / Tokyo (T-M8, 260), British Airways / Heathrow (T-M9, 223), and Korean carriers / Seoul–Incheon (T-M10, 217). The co-occurrence of strongly positive terms with service caveats in these clusters indicates that passengers who valued the core product but encountered specific shortfalls are an actionable segment for targeted recovery.

Table 4: BERTopic results - positive sentiment topics (top 10 by size).

Topic	Docs	Top keywords	Theme
T-P1	847	canada, toronto, rouge, air, westjet, ac, vancouver, montreal	Canada / Toronto / Rouge
T-P2	836	bangkok, thai, bkk, airway, flight, chiang, check, service	Bangkok / Thai / BKK
T-P3	661	qantas, sydney, brisbane, auckland, melbourne, zealand, australia, virgin	Qantas / Sydney / Brisbane
T-P4	509	doha, qatar, airway, doh, via, qr, boeing, good	Doha / Qatar / Airway
T-P5	503	lufthansa, frankfurt, munich, berlin, lh, eurowings, flight, fra	Lufthansa / Frankfurt / Munich
T-P6	458	guangzhou, southern, china, chinese, airline, service, melbourne, staff	Guangzhou / Southern / China
T-P7	407	istanbul, turkish, ist, tk, airline, good, pegasus, business	Istanbul / Turkish / IST
T-P8	378	ba, heathrow, british, london, airway, club, seat, flight	BA / Heathrow / British
T-P9	359	paris, france, cdg, air, orly, af, french, flight	Paris / France / CDG
T-P10	353	jfk, york, new, seat, food, class, flight, meal	JFK / New York

6.6 Business Risk-Factor Extraction

Synthesizing the negative and mixed themes, the framework isolates six recurrent business-risk factors, each tied to a service dimension and an evidentiary topic:

1. **Operational reliability - delays and cancellations** (T-N2, T-N7). The single most pervasive, cross-carrier complaint directly erodes trust and connection reliability.

2. **Staff service quality - rudeness and unprofessional conduct** (T-N9). A cross-carrier driver of reputational and loyalty damage.

3. **Pricing transparency - opaque ancillary fees** (T-N4) Charges for boarding-pass printing, baggage, and seat selection generate disproportionate dissatisfaction.

4. **Cabin comfort and in-flight entertainment** (T-N5) Seat space and malfunctioning IFE drive negative sentiment even on otherwise reputable carriers.

5. **Customer-service recovery - refunds and complaint handling** (T-N1, T-N8) Friction in resolving disruptions amplifies an already-negative experience.

Table 5: BERTopic results - negative sentiment topics (top 10 by size).

Topic	Docs	Top keywords	Theme
T-N1	1014	united, airline, houston, flight, newark, hour, customer, chicago	United / Airline / Houston
T-N2	817	delayed, delay, hour, time, flight, plane, minute, gate	Delayed / Delay / Hour
T-N3	683	mexico, aeromexico, city, volaris, cancun, guadalajara, interjet, tijuana	Mexico / Aeromexico / City
T-N4	664	ryanair, dublin, print, boarding, pay, euro, check, priority	Ryanair / Dublin / Print
T-N5	633	rouge, canada, air, ac, toronto, entertainment, seat, ipad	Rouge / Canada / Air
T-N6	633	etihad, abu, dhabi, airway, via, food, emirate, manchester	Etihad / Abu Dhabi
T-N7	515	delta, atlanta, time, hour, flight, delayed, fly, get	Delta / Atlanta / Time
T-N8	488	southwest, customer, time, airline, flight, fly, last, get	Southwest / Customer / Time
T-N9	482	rude, attendant, staff, asked, crew, like, people, attitude	Rude Staff / Attitude
T-N10	482	sydney, melbourne, tigerair, virgin, tiger, brisbane, australia, coast	Sydney / Melbourne / Tigerair

Table 6: BERTopic results - mixed sentiment topics (top 10 by size).

Topic	Docs	Top keywords	Theme
T-M1	459	qatar, doha, airway, via, amazing, best, service, doh	Qatar / Doha / Airway
T-M2	432	paris, france, cdg, air, french, af, orly, economy	Paris / France / CDG
T-M3	429	guangzhou, southern, china, chinese, melbourne, auckland, cabin, york	Guangzhou / Southern / China
T-M4	348	lax, sfo, icn, syd, mnl, aa, class, seat	LAX / SFO / ICN
T-M5	330	aegean, athens, greece, ath, greek, olympic, thessaloniki, santorini	Aegean / Athens / Greece
T-M6	302	klm, amsterdam, schiphol, am, dutch, via, transavia, flight	KLM / Amsterdam / Schiphol
T-M7	293	hanoi, vietnam, chi, ho, minh, vietjet, city, saigon	Hanoi / Vietnam / Ho Chi Minh
T-M8	260	tokyo, ana, narita, japan, jal, japanese, haneda, osaka	Tokyo / ANA / Narita
T-M9	223	heathrow, british, london, ba, airway, good, miami, economy	Heathrow / British / London
T-M10	217	seoul, korean, incheon, asiana, air, gimpo, korea, hyatt	Seoul / Korean / Incheon

6. **Ground and boarding processes** (T-N4, queue/gate terms) Boarding, gate, and queue management surface repeatedly as avoidable irritants.

These factors constitute the actionable output of the framework: a prioritized, evidence-backed map from passenger language to the operational levers that most influence airline business outcomes.



Figure 2: Word clouds for *positive* sentiment reviews (T-P1-T-P10): term-frequency visualization per topic cluster, in which word size is proportional to the within-cluster term frequency.

6.7 Discussion

6.7.1 Optimized Classical Models

The experimental results show that optimized MLPs and tree-based ensembles outperform fine-tuned transformer baselines, such as RoBERTa, across all datasets at a fraction of the training cost. Three factors are plausibly explained. First, this work manipulates corpora by scale, register, short snippets, and formulation review snippets with strong lexical and high-quality features (TF-IDF projected through LSA). Second, the proposed pipeline improves the feature space in which traditional models work more efficiently. Besides, transformers rely solely on loss weighting, which is deliberately excluded from SMOTE balancing. Third, transformer fine-tuning is comparatively sensitive to hyperparameters and to moderate data volumes, where the bias-variance trade-off favors lower-capacity models. This finding supports Alanazi and Li (17), who observed that an RF outperformed bespoke deep learning models for multiclass airport review prediction. It carries a practical message for production sentiment systems on review text. In addition, this well-optimized model is more accurate and dramatically cheaper than large transformers.

6.7.2 Interpreting the Sentiment Asymmetry in Topic Structure

Many negative reviews form the most coherent and separated topics, while mixed reviews are the most diverse and numerous. This is consistent with complaint dissatisfaction, which tends to be specific and attributional ('the flight was delayed','the crew was rude'), producing tight, well-defined clusters, whereas praise is generic and ambivalence is multifaceted, producing diffuse ones. Various structured and reliable feedback mitigates different risks. The emergence of carrier-agnostic negative clusters (i.e., delays and staff conduct) further indicates that these are industry-wide failure modes rather than idiosyncrasies of particular airlines, strengthening their relevance as universal strategic

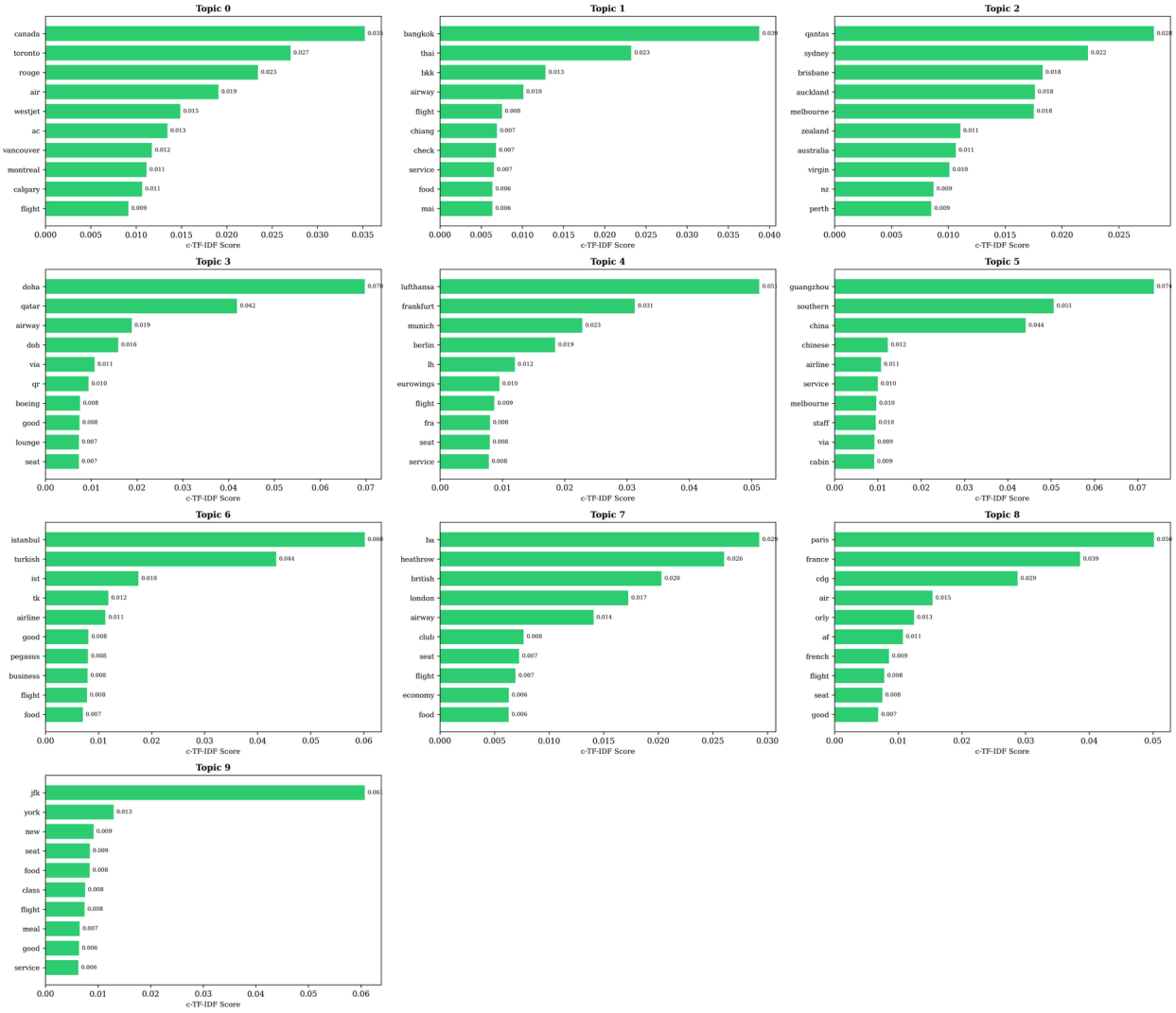


Figure 3: c-TF-IDF scores for *positive* sentiment reviews (T-P1–T-P10): the top-10 discriminative terms per topic cluster.

priorities.

6.7.3 Business and Strategic Implications for Bangladeshi Aviation

Six risk factors are extracted as a concrete agenda of this proposed model to compete in the Bangladeshi market. First, operational reliability should be the foremost investment, with padding of historical on-time data, proactive disruption management, and transparent delay communication, which directly target the largest complaint cluster. Second, staff service quality warrants structured soft-skills and de-escalation training, with feedback loops used to review monitoring and conduct targeted work to address high-frequency negative sentiments at low cost. Third, the Ryanair-style backlash suggests a clearly disclosed, predictable ancillary-pricing differentiator rather than merely a revenue line item against hidden fees. This keeps pricing transparency in a price-sensitive market. Other factors, such as cabin comfort and IFE reliability, robust refund and complaint-resolution processes, and streamlined boarding, are also considered to fulfill the agenda of the proposed model.



Figure 4: Word clouds for *negative* sentiment reviews (T-N1-T-N10): term-frequency visualization per topic cluster, in which word size is proportional to the within-cluster term frequency.

This framework verifies each recommendation against quantified evidence (topic size and sentiment) and airline authorities, and prioritizes interventions based on expected business impact rather than intuition. It also runs to monitor interventions and move the sentiment distribution periodically.

6.7.4 Comparison with Prior Work

The proposed framework compares favorably with the existing works on airline sentiment. Its best accuracy (0.9004) exceeds the performance of the hybrid VADER deep-learning model of Johari et al. (13) (74.10%), and the SMODT approach of Naresh et al. (10) (89.47%), and sits at the upper end of the 68%-98% range cataloged by Samir et al. (9), while most of those studies, explicitly engineered for class imbalance and reporting computational cost. On the other side, the BERTopic advances beyond the LDA-VADER pairing of Yu et al. (12), which produces class-specific, embedding themes with quantified coherence, diversity, and separation, and beyond aspect-based methods (17) by discovering emergent themes. The recurrence of delay-, staff-, and refund-related complaints also corroborates the qualitative findings of Yu et al. (12), lending external validity to the extracted risk factors.

6.7.5 Technical Implications

The findings carry several implications for the design of text analytics systems in service domains:

- **Benchmark pipelines before committing to transformers.** When the short-text sentiment data is analyzed, an optimized MLP and tree-based ensembles matched or exceeded a fine-tuned RoBERTa (up to 0.9004 accuracy) while training is faster ($\approx$ 9-10 s versus 376-428 s). Thus, the practitioners treat well-structured models as strong baselines rather than heavy transformer fine-tuning, especially under compute or latency constraints.

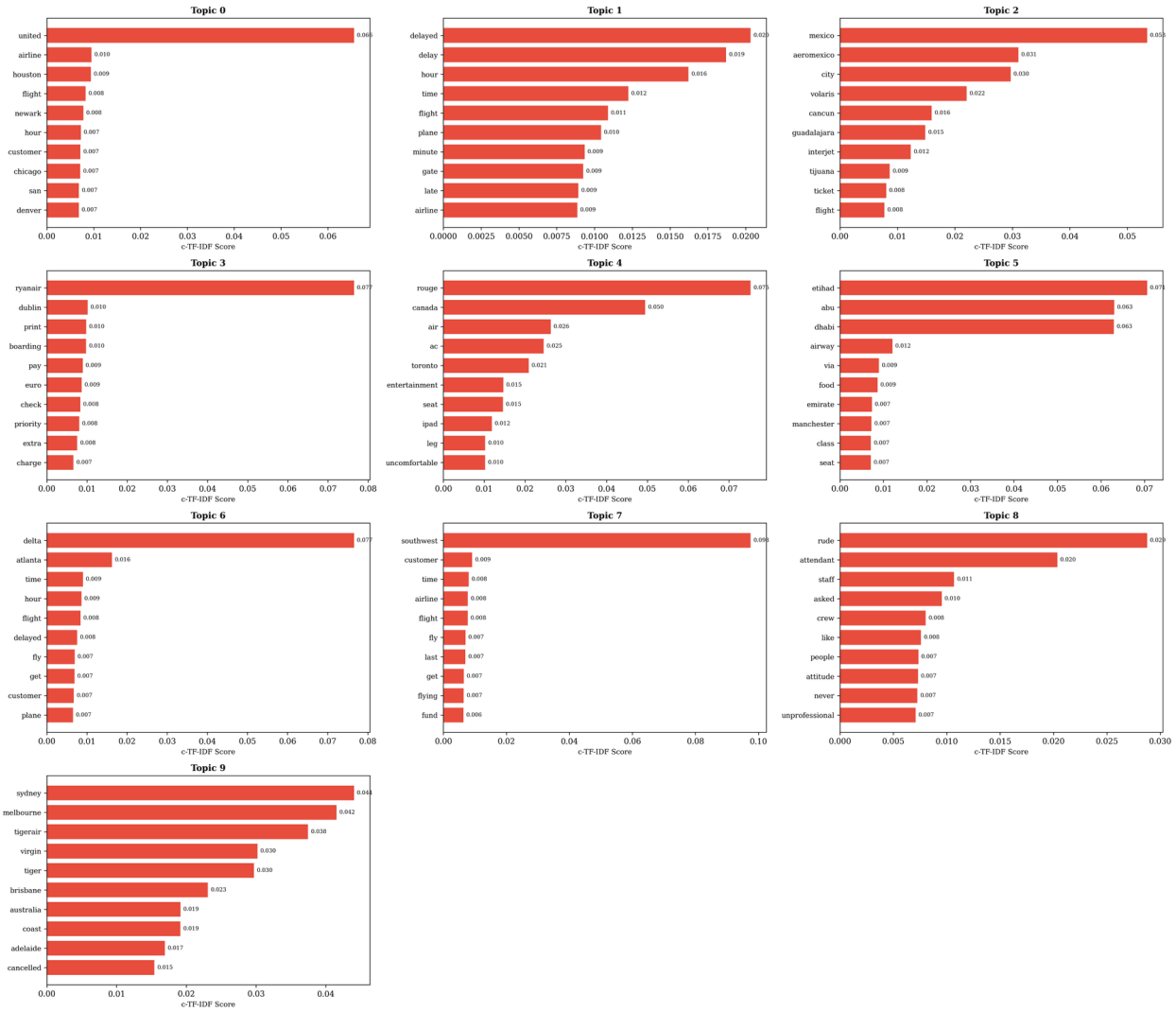


Figure 5: cTF-IDF scores for *negative* sentiment reviews (T-N1-T-N10): the top-10 discriminative terms per topic cluster.

- **Class imbalance.** The proposed pipeline applies latent-space, cluster-aware SMOTE only to numeric-feature models, while sequence and transformer models receive raw text with inverse-frequency class-weighted loss (Eq. 9). This hybrid balancing strategy is transferable to any multi-paradigm benchmark and avoids injecting semantically synthetic samples into subword tokenizers.

- **Recursive re-clustering within classes.** K-Means applies within each sentiment class, recursively splitting dominant clusters, filtering sparse clusters, and then oversampling produced a cleaner, better-conditioned feature space in which most numeric-feature models cleared 0.86 accuracy and 0.95 AUC. The procedure is model-agnostic and reusable as a general preprocessing step for imbalanced short-text corpora.

- **Conditioning topic models.** BERTopic run separately within each predicted sentiment class (see Algorithm 2). It produced well-separated topics, most coherent for the negative class ($C_V = 0.4808$, silhouette $= 0.6165$); it isolates the complaint structure, and corpus-wide modeling would blur it. This two-stage architecture is a reusable template for aspect discovery in other review



Figure 6: Word clouds for *mixed* sentiment reviews (T-M1–T-M10): term-frequency visualization per topic cluster, in which word size is proportional to the within-cluster term frequency.

domains.

- **Deployment of reusable artifacts.** It persists the fitted TF-IDF, LSA, scaler, and classifier as frozen artifacts, allowing the best model to label new, unlabelled corpora on demand and feed the topic directly, supporting reproducible and continuously re-runnable pipelines rather than one-off experiments.

6.7.6 Managerial Implications

For airline decision makers, particularly carriers competing in the price-sensitive Bangladeshi market, the framework converts unstructured feedback into a prioritized action agenda:

- **Prioritize interventions by quantified evidence.** Because the six extracted risk factors are tied to a measurable topic size and sentiment, managers can allocate scarce resources to the issues with the largest impact, such as flight delays and unprofessional staff behavior, rather than relying on intuition or anecdotes.

- **Handling cross-carrier risks as universal priorities.** The two largest negative clusters (i.e., delays and rude staff) are carrier-agnostic and tend to industry-wide failure modes. Investments in on-time performance, proactive disruption communication, structured soft-skills training, and de-escalation training for crews are the highest-leverage, lowest-regret actions for any carrier.

- **Compete on pricing transparency.** The strong backlash against opaque ancillary fees indicates that clearly disclosed, predictable pricing can serve as a differentiator in a cost-conscious market rather than merely a revenue mechanism.

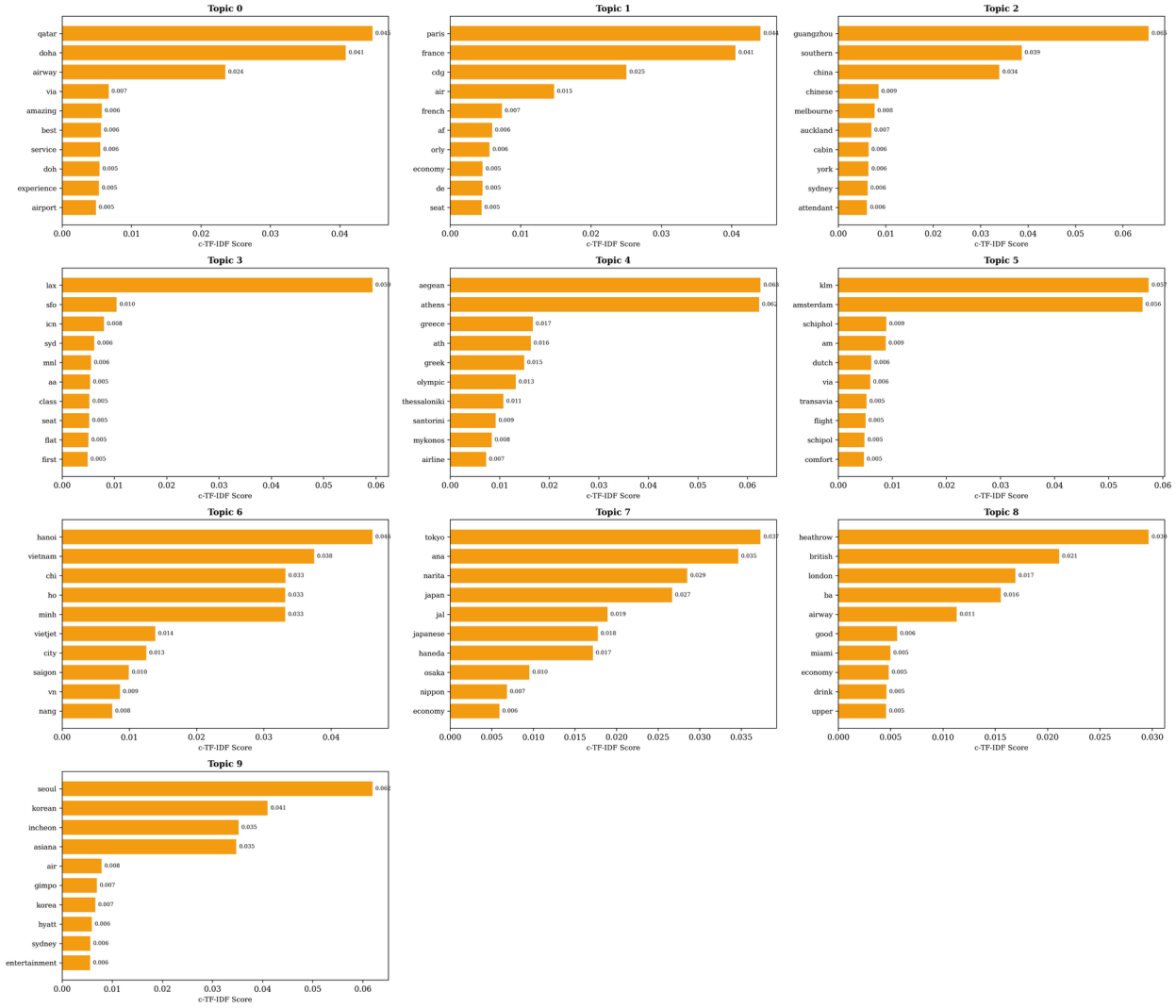


Figure 7: cTF-IDF scores for *mixed* sentiment reviews (T-M1-T-M10): the top-10 discriminative terms per topic cluster.

- **Replace costly surveys with scalable, low-cost monitoring.** The existing models are computationally inexpensive, where continuous review monitoring is feasible even for carriers with modest IT budgets, offering real-time service-quality and competitive benchmarking against rivals using only publicly available reviews.
- **Target the mixed-sentiment segment for service recovery.** Mixed reviews capture passengers who valued the core product. It encountered specific shortfalls of a well-defined, addressable segment where targeted recovery can convert ambivalence into loyalty.
- **Close the loop with a sentiment KPI.** The pipeline periodically manages to track interventions that actually shift the sentiment distribution and topic mix over time, turning the framework into an ongoing performance dashboard rather than a static study.

6.7.7 Limitations and Threats

Several limitations are associated with this work, which are given below:

- There is insufficient Bangladeshi data to analyze appropriate sentiment patterns. The demonstrated review corpus is multi-carrier and predominantly international, and Bangladeshi carriers are not separately isolated in the topic tables. Thus, the contribution to the Bangladeshi sector is principally methodological and diagnostic, a transferable framework and a set of globally validated service-risk dimensions rather than carrier-specific findings. Applying the pipeline to a dedicated corpus of Biman Bangladesh reviews, as well as reviews from other service providers such as US-Bangla, Air Astra, and NOVOAIR, is the natural next step.
- Like much of the prior literature (13), the analysis is English-centric and may under-represent Bangla and code-mixed feedback, potentially obscuring local linguistic and cultural nuance.
- SMOTE operates in the LSA space and benefits only numeric-feature models; while it means the transformer baselines are not identically balanced, so their underperformance should be read with that asymmetry in mind.
- D1 is a separate corpus from D2/D3, so the case for the optimization rests on consistently strong cross-dataset performance rather than a single controlled ablation; a same-corpus with/without-optimization ablation would further strengthen the claim.
- The absolute coherence values are moderate because they are common for short-text review data. Thus, individual topic labels should be treated as interpretive aids rather than definitive categories.
- The analysis reflects a fixed time window and does not yet model temporal drift in sentiment or the effect of interventions.

7 Conclusion

This study is designed to convert the abundant but unstructured stream of airline customer reviews into decision-grade intelligence about the business impact of service quality, with the Bangladeshi aviation market as its motivating context. To that end, it proposed a two-phase framework that unites a class-imbalance aware sentiment classifier with a sentiment-aware topic-modeling procedure. The first phase introduced a data optimization pipeline where TF-IDF, LSA, class-wise K-Means with recursive re-clustering, sparse-cluster filtering, and within-cluster SMOTE paired with class-weighted training for text models. Across three datasets, the optimized MLP delivered the best results (up to 0.9004 **accuracy,** 0.9006 **macro-F1, and** 0.9683 **AUROC**), with tree-based ensembles close behind and transformer baselines trailing at far greater computational cost, while well-engineered classical pipelines remain highly competitive for review-text sentiment. In the second phase, the best classifier is used to label a large corpus, and BERTopic is applied within each sentiment class, with negative reviews yielding the most coherent and well-separated topics (C_V = 0.4808, silhouette = 0.6165). From these themes, the framework distilled six recurrent business-risk factors, including flight delays, unprofessional staff, opaque ancillary fees, cabin comfort and IFE, refund and complaint handling, and boarding processes. Afterward, two of which (i.e, delays and staff conduct) are proven to be cross-carrier and therefore industry-wide priorities. Thus, the principal contribution is an objective, scalable, and low-cost analytical instrument that

maps passenger language to the operational levers most consequential for airline business outcomes, enabling evidence-based prioritization of service improvements.

In the future, this model will assemble and analyze a dedicated, multilingual (Bangla and code-mixed) corpus of Bangladeshi carriers to produce carrier-specific findings. Then, it will conduct a same-corpus ablation to quantify the marginal contribution of each optimization stage and incorporate aspect-level and temporal modeling to track how sentiment and risk factors evolve in response to interventions. Again, it will run the pipeline as a continuously updating monitoring dashboard for airline decision-makers.

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Appendices

A Model Comparison Results

This appendix presents the complete classification performance results for all evaluated models across the three datasets used in this study. Each table reports Accuracy, Precision, Recall, F1-score, AUROC, and training time (in seconds) for every model. Transformer-based models (BERT, RoBERTa, DistilBERT, ALBERT) and deep learning architectures (CNN, LSTM, BiLSTM, GRU) do not report training time, as their training was conducted under a different computational configuration (denoted by "–").

A.1 Details experimental results for Dataset-1 (D1)

Table 7: Model Comparison Results on Dataset-1 (D1)

Model	Accuracy	Precision	Recall	F1	AUROC	Time (s)
ET	0.8524	0.8621	0.8524	0.8520	0.9455	5.60
RF	0.8095	0.8161	0.8095	0.8085	0.9281	2.62
CNN	0.7571	0.7704	0.7571	0.7570	0.9027	–
LGBM	0.7571	0.7583	0.7571	0.7573	0.8773	0.87
XGBoost	0.7476	0.7503	0.7476	0.7483	0.8768	8.06
RoBERTa	0.7476	0.7467	0.7476	0.7421	0.8962	–
MLP	0.7286	0.7349	0.7286	0.7253	0.8977	4.71
LinearSVC	0.7286	0.7458	0.7286	0.7294	0.9141	0.46
GBC	0.7238	0.7278	0.7238	0.7241	0.8890	53.62
LSTM	0.7190	0.7606	0.7190	0.7221	0.9091	–
Ridge Classifier	0.7190	0.7343	0.7190	0.7198	0.9096	0.47
LR	0.7143	0.7244	0.7143	0.7133	0.9176	0.56
BERT	0.7143	0.7133	0.7143	0.7126	0.8806	–
GRU	0.7143	0.7296	0.7143	0.7132	0.8844	–
Bagging	0.7095	0.7136	0.7095	0.7108	0.8857	11.74
DistilBERT	0.7095	0.7060	0.7095	0.7072	0.8741	–
ALBERT	0.7000	0.6944	0.7000	0.6925	0.8576	–
SGD	0.6905	0.6888	0.6905	0.6892	0.7989	0.27
AdaBoost	0.6810	0.6774	0.6810	0.6786	0.8151	16.95
Perceptron	0.6762	0.6749	0.6762	0.6741	0.7881	0.51
BiLSTM	0.6524	0.6520	0.6524	0.6435	0.8502	–
DT	0.6238	0.6352	0.6238	0.6260	0.7179	0.48
GaussianNB	0.6143	0.6422	0.6143	0.6053	0.7129	0.10
BernoulliNB	0.6143	0.6420	0.6143	0.5901	0.8414	0.11
SVC(RBF)	0.5762	0.6217	0.5762	0.5522	0.8059	9.50

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Model	Accuracy	Precision	Recall	F1	AUROC	Time (s)
LDA	0.5286	0.5285	0.5286	0.5197	0.6423	4.61
KNN	0.3857	0.5892	0.3857	0.2885	0.6600	0.35
QDA	0.1952	0.1287	0.1952	0.1496	0.2799	3.04

A.2 Details experimental results for Dataset-2 (D2)

Table 8: Model Comparison Results on Dataset-2 (D2)

Model	Accuracy	Precision	Recall	F1	AUROC	Time (s)
MLPC	0.8923	0.8913	0.8923	0.8917	0.9612	9.02
XGBoost	0.8839	0.8814	0.8839	0.8820	0.9665	13.97
ET	0.8815	0.8839	0.8815	0.8776	0.9741	6.02
LGBM	0.8787	0.8762	0.8787	0.8768	0.9641	9.79
SVC(RBF)	0.8750	0.8722	0.8750	0.8721	0.9657	22.15
BERT	0.8630	0.8622	0.8630	0.8624	0.9548	–
RF	0.8619	0.8611	0.8619	0.8575	0.9651	31.97
DistilBERT	0.8475	0.8435	0.8475	0.8451	0.9541	–
Bagging	0.8470	0.8431	0.8470	0.8433	0.9562	83.75
RoBERTa	0.8453	0.8392	0.8453	0.8414	0.9501	–
ALBERT	0.8432	0.8419	0.8432	0.8422	0.9489	–
GBC	0.8176	0.8114	0.8176	0.8121	0.9372	251.21
LR	0.7901	0.7835	0.7901	0.7858	0.9188	0.39
LDA	0.7859	0.7839	0.7859	0.7838	0.9133	0.47
LinearSVC	0.7803	0.7676	0.7803	0.7694	0.9122	0.13
SGD	0.7663	0.7527	0.7663	0.7532	0.8988	0.90
Ridge Classifier	0.7645	0.7487	0.7645	0.7487	0.9026	0.05
CNN	0.7599	0.7797	0.7599	0.7665	0.8984	–
QDA	0.7500	0.7536	0.7500	0.7512	0.8932	1.18
DT	0.7444	0.7441	0.7444	0.7442	0.7921	3.21
AdaBoost	0.7439	0.7525	0.7439	0.7457	0.8615	46.00
BiLSTM	0.7366	0.7676	0.7366	0.7476	0.8739	–
KNN	0.7332	0.7485	0.7332	0.7363	0.8833	0.48
Perceptron	0.7318	0.7252	0.7318	0.7279	0.8642	0.13
LSTM	0.7260	0.7385	0.7260	0.7315	0.8548	–
GaussianNB	0.6926	0.7094	0.6926	0.6977	0.8482	0.03
GRU	0.6843	0.7222	0.6843	0.6985	0.8388	–
BernoulliNB	0.6819	0.6790	0.6819	0.6784	0.8388	0.04

A.3 Details experimental results for Dataset 3 (D3)

Table 9: Model Comparison Results on Dataset-3 (D3)

Model	Accuracy	Precision	Recall	F1	AUC	Time (s)
MLPC	0.9004	0.9012	0.9004	0.9006	0.9683	10.41
SVC(RBF)	0.8899	0.8876	0.8899	0.8881	0.9720	30.88
XGBoost	0.8867	0.8849	0.8867	0.8853	0.9701	16.92
LGBM	0.8853	0.8832	0.8853	0.8837	0.9676	9.21
ET	0.8791	0.8800	0.8791	0.8748	0.9705	5.79
RF	0.8683	0.8673	0.8683	0.8644	0.9605	44.65
BERT	0.8529	0.8531	0.8529	0.8529	0.9547	–
Bagging	0.8511	0.8472	0.8511	0.8479	0.9503	151.30
DistilBERT	0.8498	0.8486	0.8498	0.8490	0.9494	–
RoBERTa	0.8498	0.8485	0.8498	0.8491	0.9498	–
ALBERT	0.8486	0.8472	0.8486	0.8477	0.9490	–
GBC	0.8353	0.8295	0.8353	0.8310	0.9407	315.35
LDA	0.8126	0.8083	0.8126	0.8100	0.9249	0.31
LR	0.8122	0.8055	0.8122	0.8080	0.9300	2.02
LinearSVC	0.8025	0.7909	0.8025	0.7913	0.9252	0.17
Ridge Classifier	0.7964	0.7832	0.7964	0.7815	0.9175	0.09
SGD	0.7888	0.7775	0.7888	0.7795	0.9097	1.34
KNN	0.7849	0.7849	0.7849	0.7824	0.9065	0.46
QDA	0.7835	0.7895	0.7835	0.7857	0.9051	0.40
AdaBoost	0.7716	0.7748	0.7716	0.7720	0.8823	47.55
CNN	0.7600	0.7691	0.7600	0.7638	0.9036	–
Perceptron	0.7529	0.7515	0.7529	0.7521	0.8753	0.11
DT	0.7475	0.7504	0.7475	0.7488	0.7885	4.29
LSTM	0.7458	0.7430	0.7458	0.7444	0.8687	–
GaussianNB	0.7295	0.7334	0.7295	0.7312	0.8646	0.03
BernoulliNB	0.7237	0.7174	0.7237	0.7196	0.8659	0.04
BiLSTM	0.6886	0.7459	0.6886	0.7050	0.8546	–
GRU	0.6775	0.7216	0.6775	0.6914	0.8436	–